

Imp # Mathematical Induction

The induction means a method of defining the general statement from the validity of particular place.

It is a technique by which we can prove mathematical statement involving positive integers.

Steps Involved:

1. Inductive base; verify that $P(1)$ is true.
2. Inductive hypothesis; assume that $P(k)$ is true for an arbitrary value of k .
Inductive step;
3. Verify that $P(k+1)$ is true on the basis of inductive hypothesis.

Q. Show that: $1^2 + 2^2 + 3^2 + \dots + n^2 = \frac{n(n+1)(2n+1)}{6}$, $n \geq 1$

Solⁿ:

- Put $n=1$,

$$\text{LHS: } 1^2 = 1$$

$$\text{RHS: } \frac{1(1+1)(2+1)}{6} = \frac{1 \times 2 \times 3}{6} = 1$$

$\therefore \text{LHS} = \text{RHS}$, $P(1n)$ is true for $n=1$.

- Assuming $P(k)$ is true,

$$1^2 + 2^2 + 3^2 + \dots + k^2 = \frac{k(k+1)(2k+1)}{6} \quad \text{--- (1)}$$

Put $n=k+1$,

$$1^2 + 2^2 + 3^2 + \dots + k^2 + (k+1)^2 = \frac{(k+1)(k+2)(2k+3)}{6}$$

From eqⁿ. (1),

$$\frac{k(k+1)(2k+1)}{6} + (k+1)^2 = \frac{(k+1)(k+2)(2k+3)}{6}$$

$$\Rightarrow k(2k+1) + (k+1)6 = (k+2)(2k+3)$$

$$\Rightarrow 2k^2 + k + 6k + 6 = 2k^2 + 3k + 4k + 6$$

$$\Rightarrow 2k^2 + 7k + 6 = 2k^2 + 7k + 6 \text{ proved.}$$

Hence, $P(n)$ is also true.

Q. Show that: $1 + 3 + 3^2 + \dots + 3^{n-1} = \frac{3^n - 1}{2}$

Solⁿ:

- Put $n=1$,

$$\text{LHS: } 3^{1-1} = 3^0 = 1$$

$$\text{RHS: } \frac{3^1 - 1}{2} = \frac{2}{2} = 1$$

- Assuming $P(k)$ is true,

$$1 + 3 + 3^2 + \dots + 3^{k-1} = \frac{3^k - 1}{2} \quad \text{--- (1)}$$

- Put $n = k+1$,

$$1 + 3 + 3^2 + \dots + 3^{k-1} + 3^{(k+1)-1} = \frac{3^{k+1} - 1}{2}$$

From eqⁿ. (1),

$$\frac{3^k - 1}{2} + 3^k = \frac{3^{k+1} - 1}{2}$$

$$\Rightarrow 3^k - 1 + 2 \times 3^k = 3^{k+1} - 1$$

~~$$\Rightarrow 3^k + 3^k \cdot 3^k = 3^k \cdot 3$$~~

$$\Rightarrow 3^k + 2 \times 3^k = 3^k \cdot 3$$

$$\Rightarrow 3 \cdot 3^k = 3 \cdot 3^k \text{ (True)}$$

$\therefore P(n)$ is also true.